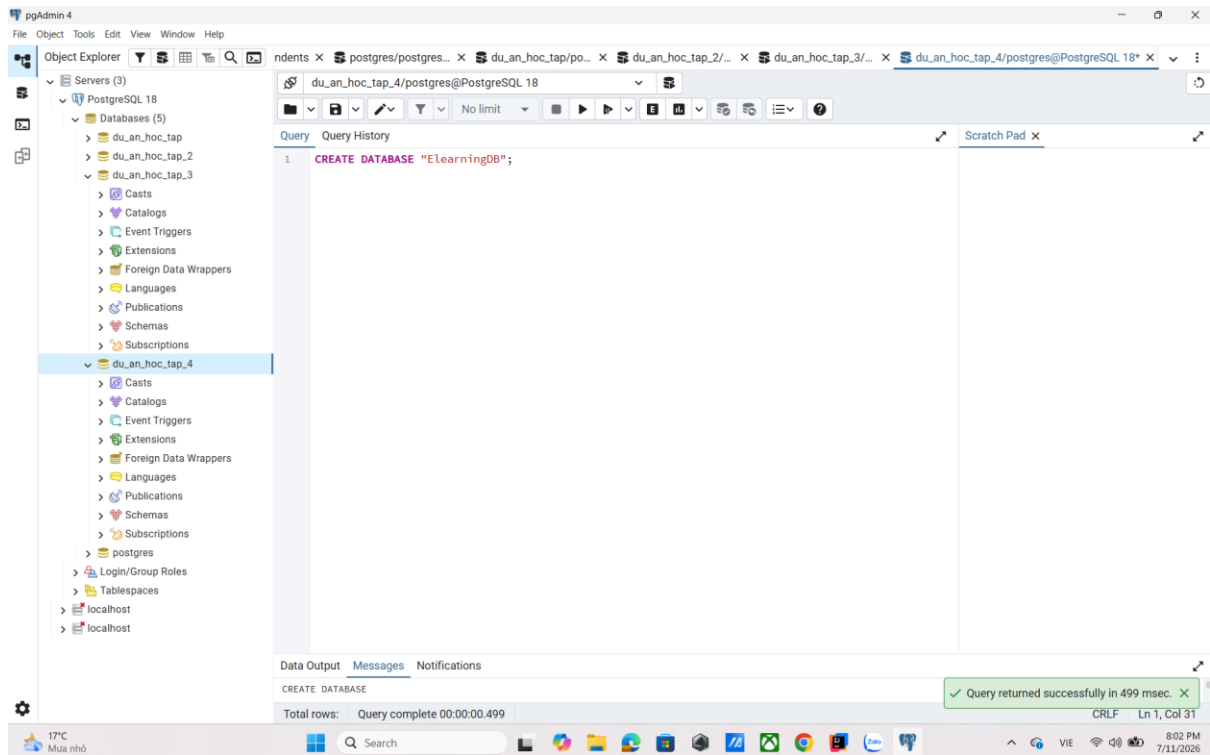


Tên: Nguyễn Hữu Ninh

Lớp: Java Backend (JB-IOC-260706)

1. Tạo database ELearningDB



2. Tạo schema và các bảng

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

- Servers (3)
 - PostgreSQL 18
 - Databases (5)
 - du_an_hoc_tap
 - du_an_hoc_tap_2
 - du_an_hoc_tap_3
 - du_an_hoc_tap_4
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - postgres
 - Login/Group Roles
 - Tablespaces
 - localhost
 - localhost

Query

```
1 CREATE DATABASE "eLearningDB";
2 -- 1. Tạo schema
3 CREATE SCHEMA elearning;
4 -- Nhóm bảng lớn (Tạo trước)
5 -- 2a. Bảng Students (Sinh viên)
6 CREATE TABLE elearning.Students (
7     student_id SERIAL PRIMARY KEY,
8     first_name VARCHAR(50) NOT NULL,
9     last_name VARCHAR(50) NOT NULL,
10    email VARCHAR(255) NOT NULL UNIQUE
11);
12 -- 2b. Bảng Instructors (Giảng viên)
13 CREATE TABLE elearning.Instructors (
14     instructor_id SERIAL PRIMARY KEY,
15     first_name VARCHAR(50) NOT NULL,
16     last_name VARCHAR(50) NOT NULL,
17     email VARCHAR(255) NOT NULL UNIQUE
18);
19 -- Nhóm bảng nhỏ (Tạo sau)
20 -- 2c. Bảng Courses (Khóa học)
21 CREATE TABLE elearning.Courses (
22     course_id SERIAL PRIMARY KEY,
23     course_name VARCHAR(100) NOT NULL,
24     instructor_id INTEGER REFERENCES elearning.Instructors(instructor_id)
25);
26 -- 2d. Bảng Enrollments (Đăng ký học)
27 CREATE TABLE elearning.Enrollments (
28     enrollment_id SERIAL PRIMARY KEY,
29     student_id INTEGER REFERENCES elearning.Students(student_id),
30     course_id INTEGER REFERENCES elearning.Courses(course_id),
31     enroll_date DATE NOT NULL
32);
33 -- 2e. Bảng Assignments (Bài tập)
34 CREATE TABLE elearning.Assignments (
35     assignment_id SERIAL PRIMARY KEY,
36     course_id INTEGER REFERENCES elearning.Courses(course_id),
37     title VARCHAR(100) NOT NULL,
38     due_date DATE NOT NULL
39);
40 -- 2f. Bảng Submissions (Nộp bài)
41 CREATE TABLE elearning.Submissions (
42     submission_id SERIAL PRIMARY KEY,
43     assignment_id INTEGER REFERENCES elearning.Assignments(assignment_id),
44     student_id INTEGER REFERENCES elearning.Students(student_id),
45     submission_date DATE NOT NULL,
46     grade NUMERIC(5,2) CHECK (grade >= 0 AND grade <= 100) -- Điểm số thực từ 0 đến 100
47);
```

Data Output Messages Notifications

CREATE TABLE

Query returned successfully in 35 msec.

Total rows: Query complete 00:00:00.035

pgAdmin 4

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Query

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45     submission_date DATE NOT NULL,
46     grade NUMERIC(5,2) CHECK (grade >= 0 AND grade <= 100) -- Điểm số thực từ 0 đến 100
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